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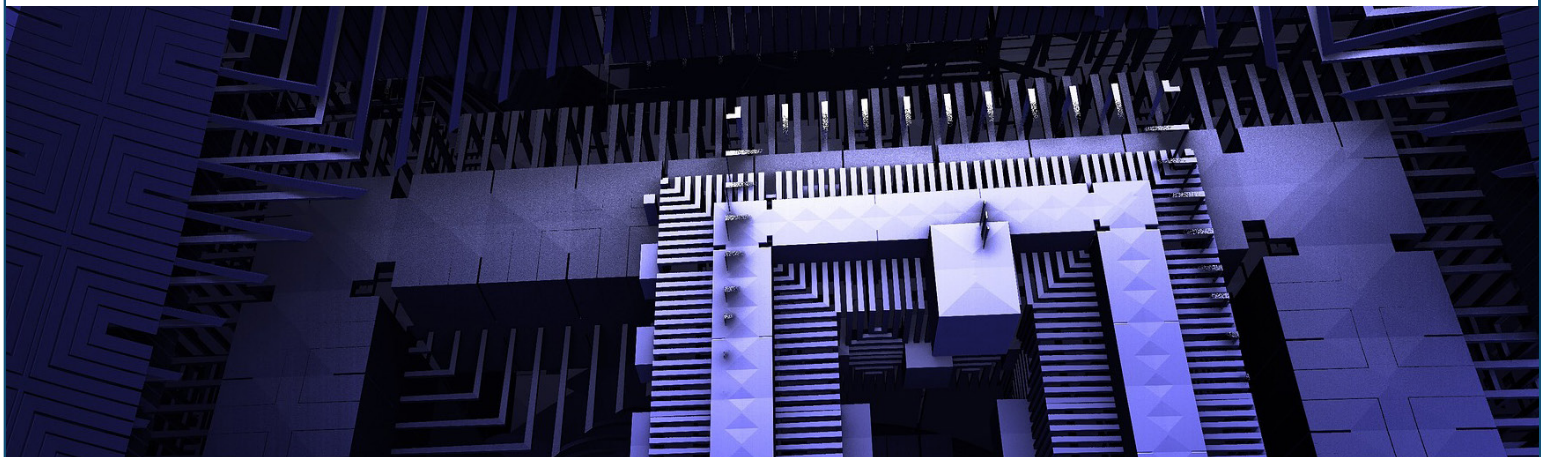

ANALYZING NATIONAL QUANTUM-COMPUTING ECOSYSTEMS IN THE INDO-PACIFIC

REPORT BY HODAN OMAAR, INFORMATION TECHNOLOGY AND INNOVATION FOUNDATION

Quantum computing is emerging as a foundational technology with implications for national security, advanced manufacturing, communications, and long-term economic competitiveness. Governments across the Indo-Pacific are investing in it heavily. This report from the National Bureau of Asian Research (NBR) examines the quantum-computing ecosystems of the United States, China, Japan, and South Korea. These four economies are not simply at different stages of development; they represent fundamentally different models of technological development—of how frontier technology gets built, financed, and brought to market. Each combines research institutions, firms, capital markets, and state policy in distinct ways, and those structural differences will shape not just who builds early prototypes but who ultimately commercializes, scales, and deploys quantum technologies at an industrial scale.

Rather than attempt to rank countries by technical milestones—an exercise complicated by divergent hardware architectures and inconsistent benchmarking—this report focuses on a more consequential question: which national ecosystems are best positioned to absorb quantum breakthroughs and translate them into sustained economic and strategic advantage?

The four ecosystems examined represent strikingly different attempts to achieve a leading position. The United States is the only ecosystem pursuing every major quantum-computing hardware architecture simultaneously, financed by a private capital market that no other country matches. China has effectively eliminated the commercial risk of quantum development by making the state both the primary funder and the primary customer. Japan

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has pivoted late but ambitiously toward owning the software layer that sits above the hardware contest, betting that middleware matters more than qubits. South Korea, candid about entering the field late, is building the institutional and manufacturing foundations to integrate and deploy quantum technologies developed elsewhere.

To examine these differences and their impacts on quantum leadership systematically, the report analyzes four dimensions of each ecosystem:

1. *Technological specialization* reveals where each country is concentrating its innovation capacity across hardware platforms and software layers.

2. *Industrial base and commercialization* concerns which ecosystem actors are doing the building and how each ecosystem moves quantum systems from laboratory to market.

3. *Trade profiles in the quantum supply chain* situate each country within global supply chains for the components, materials, and equipment that quantum hardware depends on.

4. *National strategy and government support for commercialization* address the funding structures and specific instruments governments are deploying to accelerate quantum development and commercial deployment.

These four dimensions are linked together because ecosystems are only as strong as the connections between their parts. A country can specialize in the right technologies but lack the industrial actors to build them at scale. It can have strong firms but depend on foreign supply chains for critical inputs. It can have world-class research but a thin start-up layer, or abundant private capital but fragmented policy coordination. What makes an ecosystem genuinely capable of translating quantum investment into durable advantage is not strength in any single dimension, but how those dimensions accumulate into a coherent whole. Each section of this report adds a layer to that portrait: (1) where each country's innovation is concentrated, (2) who is doing the building and how they are bringing it to market, (3) where the supply chain supports or exposes that effort, and (4) what governments are doing to sustain it. Together, the four sections present a picture of each country's structural position in the quantum race that no single metric can capture on its own.

Technological Specialization

Understanding national quantum ecosystems requires looking at the specific technological pathways they prioritize. One way to quantify these strategic priorities is to examine the revealed technological advantage (RTA). The RTA is a normalized index that measures relative specialization rather than absolute volume or technological quality. It is derived by comparing a country's share of international patent families (IPFs) in a specific subfield to its total share of IPFs across all technology domains. An RTA value of 1.0 means the country's innovation in that subfield is exactly in line with its global average performance across all sectors.[1] A value above 1.0 indicates a clear national "edge" or specialized focus, while a value below 1.0 suggests that the country is underrepresented in that subfield compared with its other innovative activities. It is important to note that the RTA is a measure of intensity relative to a country's own internal portfolio, meaning extreme values often reflect concentrated niche strategies in smaller research systems rather than absolute industrial dominance. For instance, Australia has an RTA of 12.1 in trapped-ion systems while the United States is at 1.2. This does not mean that Australia is ten times as advanced; rather; it means that, relative to its own total patenting output across all technology areas, Australia is ten times as focused on trapped-ion technology as the United States is relative to its own total output.

Figure 3.3.5 in the 2025 report *Mapping the Global Quantum Ecosystem*, jointly published by the

European Patent Office (EPO) and Organisation for Economic Co-operation and Development (OECD), presents RTA values based on aggregated patent activity from 2005 to 2024 across major quantum subfields.[2] The following analysis uses these values to profile the distinct strategic priorities of the United States, Japan, China, and South Korea.

This NBR report is primarily concerned with quantum computing. The RTA analysis that follows, however, covers communications and sensing as well. This is not simply a matter of comprehensiveness but reflects the zero-sum nature of RTA as an analytical tool. Because RTA measures a country's share of patents in a specific field relative to its share across all fields, it acts as a map of revealed preference and scientific opportunity cost. A country's computing bets look different when viewed alongside its investments in sensing and communications. High specialization in one area often signals where an industrial base is concentrating its finite research and development capacity. By looking at all three subfields, we can distinguish between a broad-front leader that maintains intensity across the entire quantum stack and a niche specialist that has strategically over-indexed on a single domain.

Quantum computing is analyzed through five main layers of the stack. *Physical realizations* refers to the actual hardware platforms used to build quantum processors, including superconducting circuits, trapped ions, spin-based systems, and quantum optics. *Models* covers the conceptual frameworks for processing quantum information; these are divided between simulation models, which mimic complex physical or chemical processes, and non-simulation models, which tackle abstract computational problems like optimization and cryptography. *Algorithms* refers to the specific procedures designed to run on quantum hardware to solve defined problems. *Error correction* covers the techniques used to detect and fix mistakes in imperfect quantum systems—a critical layer for scaling toward reliable, commercially viable machines. *Programming* covers the software languages and tools used to write and execute quantum programs. Together, these layers span from the physical hardware at the bottom to the application-facing software at the top.

Quantum communication comprises four main subfields. *Optical realizations* covers technical adaptations to improve stability, efficiency, and error rates in transmitting quantum information using optical systems. *Quantum memories* refers to devices capable of storing quantum states with long coherence times, a capability essential for building long-distance networks, such as satellite-based systems. *Quantum key distribution* is the most commercially mature subfield, focused on achieving secure communication through quantum-encrypted keys. *Information networks* covers the higher-level architectures—including routing and connectivity—that link quantum devices like computers and sensors to form the backbone of a future quantum internet.

Quantum sensing comprises five main subfields, each defined by the physical phenomenon being measured to surpass classical sensitivity limits. *Gravitation, rotation, and acceleration sensing* uses quantum effects to detect motion and gravitational forces with extreme precision, with applications in inertial navigation and geophysical surveying. *Time measurement* covers quantum-enhanced atomic clocks and frequency standards, which underpin GPS precision, financial transaction timing, and telecommunications synchronization. *Magnetic field measurement* uses quantum sensors to detect minute magnetic variations, with applications in medical imaging and subsurface detection. *Chemical detection* identifies specific molecules or substances with high sensitivity, relevant to environmental monitoring and industrial analysis. *Imaging* covers quantum-enhanced approaches—such as ghost imaging—to capture visual or physical information at resolutions or sensitivities beyond what classical sensors can achieve.

United States. In quantum computing, the United States shows a specialized edge across every major hardware modality and nearly every layer of the quantum-computing stack. It maintains RTA values of 1.0 or higher in superconducting (1.1), trapped-ion (1.2), spin-based (1.1), and quantum optics (1.0) systems. This breadth means actors in the United States are not betting on a single architecture but preserving strategic optionality across competing computing pathways. It also shows clear specialization in error correction (1.3) and programming (1.2). Strength in these layers indicates the U.S. ecosystem is building the operating infrastructure required to scale commercially viable machines. By contrast, the United States' relative under-specialization in algorithms (0.8) may indicate that algorithmic breakthroughs are still concentrated in academia rather than patent-heavy industry, or that the United States is betting that if it builds the most reliable machines and programming languages, it can simply import application-layer innovation from the rest of the world.

In contrast to its broad-based approach to computing, the United States' strategy in quantum communication is narrow, with RTA values below the parity mark for information networks (0.7) and optical realizations (0.7). This distribution suggests a strategic emphasis on long-term technical challenges over immediate market infrastructure. However, the U.S. ecosystem has a specialization in quantum memories (1.7). By focusing on these essential storage devices, which are critical for extending the range of quantum signals, the United States appears to be prioritizing the foundational hardware required for a future global quantum internet rather than the current generation of terrestrial networking.

The U.S. profile in quantum sensing is defined by concentrated strengths in high-precision navigation and advanced imaging. The primary focus is clearly on magnetic fields (1.4), which stands as the nation's most significant specialized edge in the sensing category. This strength is tightly coupled with a focus on imaging (1.3) and gravitation, rotation, and acceleration (1.1). Together, these data points suggest an ecosystem that is optimized for precision aerospace, defense, and specialized industrial navigation where extreme sensitivity is a prerequisite. There is a notable lack of focus in time measurement (0.7)—a field where Japan holds a massive specialization.

Japan. Japan's quantum-computing ecosystem is defined by a solid position in superconducting hardware (1.1), a field where Japan has a long-standing industrial research heritage. In contrast, its RTA values are significantly lower in trapped ion (0.3) and quantum optics (0.4), reflecting a narrower hardware footprint relative to more diversified ecosystems. Japan shows a striking specialization in algorithms (1.6), far outpacing other major players, which indicates a strong emphasis on computational efficiency and software-layer innovation as core sources of advantage. This may suggest that the Japanese ecosystem is betting heavily on the intersection of superconducting circuits and high-level algorithmic efficiency rather than a diversified hardware approach.

In quantum communication, Japan's specialization in optical realizations (1.2) and quantum key distribution (1.1) suggests a strategy built on immediate industrial utility. While the United States appears to be focusing on long-distance quantum memories that remain largely in the research phase, Japan may be leveraging the manufacturing power of its electronics giants, such as Toshiba and NEC, to produce the actual physical hardware used in secure networks today. This indicates that the Japanese ecosystem is potentially geared toward becoming a global factory for the components of secure communication, possibly prioritizing the commercial production of optical fibers and encryption hardware over more abstract networking theories.

Japan's profile in quantum sensing is dominated by an extraordinary specialization in time measurement (3.5), which stands as the country's highest RTA value across the entire quantum stack. This massive outlier indicates that Japan has concentrated its sensing innovation almost exclusively on the development of ultra-precise atomic clocks and frequency standards. Outside of this niche and a modest presence in magnetic field measurement (1.1), Japan is significantly underrepresented in other sensing subfields. This points to a highly surgical strategy: Japan is not attempting to lead in general-purpose sensing but is instead aiming for global dominance in the timing infrastructure that underpins everything from satellite navigation to high-frequency financial trading.

China. China maintains solid positions in superconducting hardware (1.1), suggesting that it is building practical capabilities around a proven architecture. However, its hardware footprint is narrower than that of the United States, with significantly lower RTA values in trapped-ion (0.6) and spin-based (0.5) systems. This points to an approach focused on concentration rather than diversification across multiple qubit pathways. China shows strong specialization in non-simulation models (1.6). In quantum computing, simulation models are typically used to mimic physical systems like chemicals or materials; by contrast, China's focus on non-simulation models indicates a pivot toward general-purpose computational frameworks. This specialization suggests that its ecosystem is especially invested in the math-heavy side of quantum—specifically the logical structures required to solve abstract classes of problems like cryptography and complex optimization, rather than just simulating physical nature. China's specialized edge in error correction (1.3) may indicate a focus on stabilizing and managing quantum information in imperfect devices. Taken together, the pattern suggests a targeted strategy: pairing strength in theoretical computing models and error management with a primary bet on superconducting circuits to drive scalable progress.

In quantum communication, China's specialization in optical realizations (1.2) and a parity-level focus on quantum key distribution (1.0) reflects an ecosystem geared toward the large-scale deployment of current-generation secure network infrastructure. Again, unlike the United States' focus on long-distance research hardware, China appears to be prioritizing the immediate industrialization of quantum-secured transmission.

The most striking outlier in the entire Chinese profile is an RTA of 3.1 in chemical detection. This massive specialization indicates a strategic national pivot toward using quantum sensors for industrial, environmental, or materials science applications. By focusing so heavily on this specific niche, the Chinese ecosystem appears to be seeking a first-mover advantage in integrating quantum sensing directly into its existing, highly competitive manufacturing and chemical industrial base, while maintaining a steady but less diversified presence in imaging (1.0).

South Korea. South Korea is significantly underrepresented across the quantum-computing subsector. In every computing category measured—from physical hardware (0.2–0.3) to programming (0.4) and algorithms (0.4)—South Korea's RTA remains well below the 1.0 parity mark. This highlights a current strategy that is heavily weighted toward quantum information networks (2.9) and optical communication (1.0) rather than the processing of information itself. This approach may mean that South Korea is currently prioritizing the secure transmission of data over building quantum computers.

Instead, it has funneled its innovation capacity almost exclusively into the architectural layer of the stack, holding a world-leading RTA of 2.9 in information networks, the highest value for this subfield not only among the four ecosystems in this analysis but across all major players. This focus reflects a total commitment to the protocols and routing systems required for a future quantum internet, likely driven by a national strategy to become the global hub for quantum data transmission rather than the manufacturer of the devices themselves.

This networking-first strategy is further evidenced by South Korea's striking lack of focus in the sensing domain, where it is almost entirely absent from high-precision measurement innovation. The country holds the lowest RTA values among major players for time measurement (0.1) and magnetic fields (0.0). The only area where it approaches specialization is in chemical detection (1.6), which may serve as a specialized industrial application for the domestic tech sector.

Industrial Base and Implications for Commercialization

A nation's ability to lead in the quantum era depends not only on the technological pathways it prioritizes, but on its domestic industrial base: the specific mix of firms, research institutions, and labs capable of building and scaling those technologies. A country's industrial base defines the underlying set of actors and capabilities that make the process of commercialization—i.e., bringing technologies to market—possible.

Within any national quantum industry, two distinct dimensions of industrial organization are particularly critical for shaping the trajectory and character of a country's development. The first is the balance between core and non-core firms, a ratio that defines an industry's "quantum intensity" and reveals how structurally committed the commercial sector is to achieving specific technological breakthroughs. Core or pure-play firms are companies whose primary mission and survival are inextricably linked to quantum-technology development. Because these firms lack the safety net of legacy revenue, they are structurally incentivized to pursue high-risk, high-reward hardware or software pathways with extreme aggression.

In contrast, non-core or partial-play firms participate in the quantum sector while currently deriving the majority of their revenue from established markets. This group generally consists of two types of actors: diversified incumbents such as large platform companies, defense primes, and semiconductor giants that treat quantum as a strategic hedge; and specialized enabling suppliers, such as photonics, cryogenics, and advanced materials that serve quantum needs as well as the wider precision-technology market. Data from a Quantum Economic Development Consortium (QED-C) report highlights the scale of this divide, noting that while there are approximately 513 core quantum companies worldwide, nearly 6,000 non-core organizations are currently engaged in quantum-related activity.[3]

The strategic implication of this balance is significant. A higher density of core firms typically signals an ecosystem defined by intense architectural experimentation and competitive pressure. Conversely, an industrial base dominated by non-core firms suggests an ecosystem where quantum activity is more deeply embedded within diversified portfolios, often leading to a greater emphasis on system integration, software layers, and industrial optimization within existing supply chains.

The second important dimension of industrial organization is the distribution of start-ups, established firms, and public research organizations (PROs). Data from Figure 3.4.3 in the EPO and OECD's 2025 report *Mapping the Global Quantum Ecosystem* shows how patenting activity in quantum computing is naturally divided among start-ups, established companies, and public research organizations from 2005 to 2024. There is a clear functional split: start-ups account for a disproportionate share of activity in physical hardware modalities, such as trapped ion and quantum optics.

Because start-ups are built to take high-stakes architectural risks, a high density of these firms explains why a national ecosystem may lead in hardware breakthroughs. In contrast, established companies concentrate more heavily on the programming, algorithms, and error correction layers. This suggests that while start-ups drive the moonshot hardware experimentation, larger firms are the primary drivers of the software infrastructure and system reliability required for commercial use. Consequently, an ecosystem's organizational profile can help contextualize its specific technological strengths: a start-up-heavy base is positioned for hardware disruption, while an incumbent-heavy base is positioned for industrial deployment.

*United State*s. The United States possesses the world's largest and most structurally diversified quantum industrial base. It has more than 300 quantum-active organizations, roughly half of which are core, pure-play quantum firms. This group spans multiple hardware architectures, including trapped-ion leaders such as IonQ, photonic firms such as PsiQuantum, and superconducting developers such as Rigetti Computing, as well as software-focused companies such as Quantum Computing Inc. The remaining organizations fall into the non-core category. These include large platform companies such as Google, IBM, and Microsoft; defense primes such as Lockheed Martin and Northrop Grumman; semiconductor manufacturers such as Intel and Nvidia; and specialized component suppliers such as Coherent in photonics and Lake Shore Cryotronics in cryogenics. This composition produces an ecosystem with both high quantum intensity and significant architectural breadth, as reflected in the RTA metrics above.

The U.S. industrial base is also uniquely start-up heavy. Approximately 85% of dedicated U.S. quantum firms are early-stage, venture-backed start-ups rather than state-owned enterprises or established industrial champions. This reflects the central role of private capital in financing frontier technologies in the United States. U.S.-based firms capture nearly 60% of all global private quantum funding, giving start-ups access to capital at a scale rarely seen elsewhere. Even as the total number of global funding rounds has declined from its 2022 peak, the average deal size in the United States has increased, with investors concentrating capital into larger growth-stage financings rather than smaller seed rounds. While international start-ups often rely on fragmented public grants or modest early-stage funding, U.S. firms such as PsiQuantum and Quantinuum have secured single investment rounds ranging from $500 million to $1 billion.

The result is a long tail of highly capitalized start-ups capable of pursuing capital-intensive hardware pathways in parallel. This financing structure reinforces the experimentation visible in U.S. RTA data: multiple architectures are advanced simultaneously rather than funneled through a single state-directed pathway.

A primary benefit of the United States having dominant conventional computing giants is the unique ability of these firms to manufacture market demand. A key way they do this is by leveraging their existing cloud ecosystems to act as the commercial on-ramp for the entire industry. IBM, Google, and Amazon Web Services have packaged quantum hardware into pay-per-use platforms that allow enterprise customers—in financial services, pharmaceuticals, logistics, and defense—to experiment with quantum systems without owning hardware outright. This lowers the barrier to entry enough that commercial engagement can begin well before the technology reaches full maturity, generating the early user base and application feedback that eventually justify larger enterprise commitments. The results are beginning to solidify: HSBC reported using IBM's Heron quantum computer to improve bond trading predictions by 34% over classical computing alone.[4]

The high density of pure-play start-ups provides a complementary potential advantage. While computing giants build a commercial on-ramp, start-ups are structurally positioned to pressure-test diverse near-term applications—quantum-safe cryptography, quantum sensing for medical diagnostics, optimization tools for logistics—in real markets with real customers. If that experimentation is sustained, it does more than generate early revenue; it builds the buyer sophistication and operational knowledge that will matter when more powerful systems arrive. A primary example is Quantinuum. Its Helios system launched with a group of early industry users, including Amgen, BMW, JPMorgan Chase, and SoftBank, running applied experiments in areas ranging from drug discovery and materials science to financial analytics and battery development, giving firms hands-on experience with quantum workflows as systems advance through successive hardware generations. By the time fault-tolerant machines are ready, the United States could have spent a decade cultivating enterprise customers who already know how to integrate quantum tools into their workflows—a structural head start that no government program can easily replicate.[5]

Japan. Japan's quantum industrial base is relatively small and anchored by established, multi-sector incumbents rather than a large population of pure-play firms. Its non-core actors include major information and communications technology (ICT) and electronics companies such as Fujitsu, NEC, NTT, Hitachi, and Mitsubishi. For these firms, quantum computing sits inside broader portfolios in conventional supercomputing, telecommunications, and advanced electronics, which shapes both the pace and the type of quantum development that is commercially prioritized.

This composition helps contextualize Japan's technological profile. Because quantum work is concentrated inside large corporate R&D systems, the country's approach tends to emphasize areas where incumbents can build an advantage through engineering depth and computational efficiency rather than wide-ranging architectural exploration. That pattern aligns with Japan's RTA strengths in algorithms and superconducting hardware, as well as with its comparatively limited presence in alternative modalities such as trapped ion and quantum optics.

Japan also shows comparatively limited start-up density. As a 2023 RAND report notes, much of the country's quantum R&D is conducted within large, established corporations. A start-up layer does exist and is growing, including firms such as OptQC and QunaSys, but these companies typically operate in closer orbit to corporate partners and public research infrastructure than the independently capitalized, venture-scale model that defines the U.S. long tail.[6]

Taken together, Japan's industrial base is best characterized as incumbent-led and coordination-intensive. It is structurally well suited to reliability, integration, and application-facing progress built on corporate engineering capacity, but is less oriented toward the broad, parallel hardware experimentation produced by a large venture-backed start-up population.

An implication of Japan's industrial base is that commercialization is treated as a targeted co-development process. While the U.S. approach leverages start-ups to find niche buyers, Japan's quantum start-ups and research institutions actively seek out industrial partners—in automotive, electronics, and precision engineering—to develop tools around problems those partners are already working on and already have budgets for.

QunaSys, a quantum software start-up, illustrates this concretely. By co-developing algorithms specifically for Toyota's battery materials research, QunaSys builds tools tailored to a client's specific, high-value bottleneck. This separates Japan's demand-creation model from that of the United States: the developer and the industrial customer are working on the same problem together, ensuring the buyer is operationally invested in the outcome before a final product even exists. This is demand creation through depth rather than breadth—focusing on a few sophisticated buyers motivated by real-world industrial hurdles.

More broadly, Japan seems to be focused on commercializing the software layer, reflecting a candid reading of its own constraints. Without the venture-backed pure-play firms needed to compete across multiple hardware architectures simultaneously, Japan cannot rely on hardware sales as its primary commercial engine. The Ministry of Economy, Trade and Industry (METI) is supporting efforts by telecom giant KDDI and optimization start-up Jij to develop a quantum operating system—the software interface that sits between hardware and applications. No company has yet established a widely adopted quantum operating system, and Japan is trying to define that standard before hardware architectures consolidate. If successful, this would not just create a product; it would make Japan a leader in the layer through which all quantum hardware, including hardware built elsewhere, gets used.

China. China's quantum industrial base is smaller in firm count than that of the United States and is characterized by a lower density of core, pure-play firms. While thousands of Chinese institutions participate in quantum-related research or adjacent technologies, the number of dedicated quantum-computing companies is limited—estimated at roughly 14–30 start-ups. The ecosystem instead centers on a small group of state-aligned firms, such as Origin Quantum (superconducting systems), which function as the primary commercialization vehicles for breakthroughs emerging from national research hubs such as the National Laboratory for Quantum Information Science in Hefei.[7]

On the surface, China's recent activity looks like it could be start-up–heavy, but the underlying trend is actually one of state consolidation. Between late 2023 and early 2024, Chinese tech giants Alibaba and Baidu shut down their independent quantum-research labs and donated their equipment to state-backed institutions such as the Beijing Academy of Quantum Information Sciences and Zhejiang University. This mass exit of the "big tech" platform players effectively shrank the pool of independent industrial actors, moving high-end talent and hardware back under direct government oversight.

According to a 2025 report from the U.S.-China Economic and Security Review Commission, the sudden closure of private quantum labs in China was not a purely commercial decision but a strategic realignment "reportedly under government pressure to centralize control."[8] The commission suggests that private-sector efforts often lagged behind the cutting-edge progress made at state-funded institutions like the University of Science and Technology of China. Facing the high R&D costs of a "quantum winter" (i.e., a period of cyclic declines in research funding and innovation) and a cooling national economy, firms such as Alibaba and Baidu pivoted toward generative artificial intelligence (AI), where commercial returns are more immediate, while the state moved to eliminate wasteful duplication of quantum efforts. By redirecting personnel and hardware into state-linked entities, the Chinese Communist Party has ensured that all quantum R&D is directly subordinated to its strategic and military objectives. Furthermore, the report identifies U.S. policy as a primary catalyst for this restructuring. By moving programs into state labs, China can better insulate its supply chain from U.S. export controls on semiconductors and shield sensitive research from foreign influence, consolidating scarce components for the most promising national projects.[9]

In this environment, the start-ups that remain in China are not really start-ups in the sense of independently capitalized, venture-backed gambles. Instead, they often function as commercial appendages of the state. Many are spin-offs from government-funded universities or labs that rely on state-directed procurement rather than market competition.[10] This creates an industrial base that is structurally optimized for unified scaling and national missions but lacks the long tail of diverse, independent firms that drive parallel hardware experimentation in the United States.

The primary implication of China's consolidated industrial base is that commercialization is driven by strategic mandate rather than market demand. By thinning out its private layer, the state has removed the need for quantum technology to prove its immediate commercial value to shareholders. Instead, China uses its state-owned enterprises as a guaranteed market. For example, Origin Quantum's latest quantum computer was integrated into the national supercomputer network in April 2024, a grid of shared computing facilities used by universities, state enterprises, and government agencies across the country. Inclusion meant immediate, guaranteed usage at scale without Origin Quantum needing to find, pitch, or convert a single customer.[11]

South Korea. South Korea's quantum industrial base is relatively small in firm count and heavily weighted toward diversified incumbents rather than dedicated pure-play firms. Large chaebol groups—such as Samsung, SK Telecom, and LG—anchor much of the country's quantum activity. However, unlike incumbents such as IBM or Google, whose core businesses are rooted in large-scale computing infrastructure and cloud systems, South Korean conglomerates are structurally centered on semiconductor fabrication, telecommunications networks, and device manufacturing.

This industrial profile shapes maneuverability. Firms built around chip production and network integration can more naturally extend into quantum sensing and communications. Developing full-stack quantum-computing architectures, by contrast, requires sustained architectural risk-taking and deep software-hardware co-design capabilities that are more closely aligned with computing-centric firms. As a result, South Korea's ecosystem is more organically positioned for sensing, networking, and fabrication than for leading independently designed quantum-computing platforms.

Start-up density is correspondingly limited. Only a small number of dedicated quantum start-ups, such as Qunova Computing, SDT, and QSIMPLUS, operate within the ecosystem. South Korea's thinner start-up layer means fewer independent firms are positioned to undertake high-risk hardware experimentation across multiple qubit modalities.[12]

South Korea's industrial base is incumbent-centered and manufacturing-oriented. It is structurally positioned to contribute to large-scale fabrication and integration within the quantum supply chain but currently lacks the dense start-up layer associated with broad architectural experimentation. The result is a commercialization ecosystem oriented around what South Korea's largest incumbents already do. That concentration may produce genuine near-term commercial traction in those verticals. It also means, however, that the ecosystem generates few independent signals about where quantum might create value outside them. Without a thicker layer of start-ups pursuing diverse applications on their own terms, South Korea is less likely to stumble into the unexpected use cases that have historically defined new technology markets.

Trade Profiles in the Quantum Supply Chain

Trade profiles reveal where countries are structurally embedded in global value chains. In the context of quantum-relevant goods, three dimensions are particularly important: export value, trade specialization, and import value.

Export value captures scale. It shows which economies dominate global supply in absolute terms and therefore occupy central roles in production networks. Countries with large export values shape availability, pricing dynamics, and potential bottlenecks. Import value, by contrast, captures exposure. It reveals where countries rely on foreign suppliers for critical inputs and therefore where vulnerabilities may lie. Large import values in specific product categories can signal dependence, even for technologically advanced economies.

Trade specialization, which is measured through revealed comparative advantage (RCA), captures something different: intensity. It shows whether quantum-relevant goods represent a disproportionately large share of a country's overall trade profile. A country may export large volumes simply because it is large; specialization indicates structural focus. High RCA values suggest deeper industrial concentration and greater embeddedness in that segment of the value chain, even if total trade values are smaller.

Distinguishing between equipment and raw materials is also critical. Equipment goods, such as integrated circuits, processors, and fabrication tools, tend to reflect advanced manufacturing capabilities and participation in downstream, higher-value segments of the quantum ecosystem. Raw materials—such as specialty chemicals, aluminum oxide, rare earths, or isotopes—reflect upstream positioning and control over foundational inputs. The two segments carry different strategic implications: equipment trade highlights technological capacity and manufacturing integration, while raw materials trade highlights resource endowments, concentration risks, and potential chokepoints.

Figures 7.1.2, 7.2.1, and 7.1.3 in the EPO and OECD's report *Mapping the Global Quantum Ecosystem* present values for export volume, trade specialization (RCA), and import volume, respectively, across quantum-relevant equipment and raw material categories using the international harmonized system trade classification system known as HS-6.

One limitation of this approach is worth highlighting. Value-based trade metrics capture industrial scale well but can obscure some strategic chokepoints. Helium-3 is the clearest example: this rare isotope is essential for cooling superconducting quantum computers, but its total dollar value on the global market is minuscule compared with bulk industrial exports like aluminum. Consequently, helium-3 does not appear as a significant import or export when looking at values alone.

Still, these metrics are a valuable proxy for understanding where a country is building meaningful industrial capacity to manufacture and supply quantum-relevant goods and where it is dependent on others for critical inputs. The following sections analyze these metrics for the United States, Japan, China, and South Korea, exploring how their individual industrial substrates translate into specific trade strengths and strategic vulnerabilities.

United States. The United States participates at scale in quantum-relevant trade, but it is not disproportionately focused on these goods. It ranks among the top exporters of quantum-relevant equipment—particularly processors and fabrication equipment—though it trails Taiwan, China, and South Korea in absolute export value. In raw materials, it also sits within the top five globally, with exports concentrated in foundry binders and specialty chemicals rather than mineral-based inputs. However, its RCA remains below 1.0 in both equipment and raw materials, meaning that quantum-relevant goods account for a smaller share of U.S. exports than they do on average globally. These goods are part of a large and diversified export base rather than a defining trade pillar.

At the same time, the United States is the world's largest importer of quantum-relevant raw materials and one of the largest importers of equipment. A significant portion of this demand includes static converters—power-management components essential for delivering the precise, stable electrical control required in quantum hardware systems.

This import intensity reflects a broader pattern: domestic demand for advanced inputs exceeds domestic supply. Overall, the U.S. trade profile is defined by scale and strong absorptive capacity, yet with continued reliance on global supply chains for key upstream materials and specialized manufacturing components.

Japan. Japan's trade profile is defined less by sheer volume and more by concentration in high-precision segments of the quantum value chain. While it does not match China or Taiwan in total export value, Japan is a major exporter of high-end equipment, including fabrication tools, lasers, photonics components, and cryogenic systems that are critical for the development of quantum hardware.

This positioning is reinforced by trade specialization. Japan's RCA exceeds 1.0 in both equipment and raw materials, indicating that quantum-relevant goods constitute a larger share of the country's exports than the global average. In raw materials, Japan shows particularly strong concentration in high-purity chemical inputs such as oxometallic salts—refined materials essential for advanced semiconductor and quantum manufacturing processes. This specialization likely reflects the structure of Japan's industrial base: long-standing capabilities in precision chemical engineering and close integration with semiconductor manufacturing have produced firms that compete at the high-purity, high-reliability end of the market rather than at mass scale.[13]

Japan's import profile reinforces this specialization pattern. While the country imports raw materials to support manufacturing, it shows comparatively lower reliance on foreign-made fabrication equipment than several peer economies. This aligns with its export strength in high-precision tools and refined inputs. In other words, Japan is not just exporting specialized components; it is also producing much of the advanced equipment needed to support those industries domestically.

The overall pattern is one of concentrated capability in critical segments of the value chain. Although Japan does not dominate global trade volumes, it holds structurally strong positions in specialized equipment and high-purity materials that are essential to advanced semiconductor and quantum manufacturing.

China. China's trade profile is defined by scale. It is the largest exporter of quantum-relevant raw materials by value, driven primarily by foundry binders, specialty chemicals, and aluminum. It also ranks second globally in equipment exports, with flows concentrated in processors, integrated circuits, static converters, and electrical connectors.

However, this scale does not translate into structural specialization. China's RCA in raw materials sits around parity, and its RCA in equipment remains below 1.0. This indicates that quantum-relevant goods account for a share of the country's exports roughly proportional to the size of its broader manufacturing economy. In other words, China's dominance reflects breadth and industrial scale rather than concentrated trade focus in specific quantum-relevant segments.

At the same time, China is the world's largest importer of equipment, particularly in processors and fabrication equipment. This massive import volume, coupled with a sub-1.0 RCA in equipment exports, highlights a critical vulnerability: China is a high-volume assembler and material supplier, but it remains trade-dependent on more specialized neighbors for the high-end precision tools it cannot yet produce with a comparative advantage.

South Korea. South Korea's trade profile is more concentrated and equipment-focused than that of either the United States or China. It ranks among the leading exporters of quantum-relevant equipment, with exports heavily centered on processors and integrated circuits—segments closely tied to its strength in semiconductor manufacturing. In raw materials, South Korea also maintains meaningful export volumes, particularly in refined chemical inputs.

Unlike larger economies, South Korea exhibits clear trade specialization. Its RCA exceeds 1.0 in equipment and also stands above parity in several material categories. This indicates that quantum-relevant goods account for a larger share of the country's exports than they do on average globally, reflecting a more concentrated trade structure.

South Korea's import profile complements this pattern. The country imports advanced fabrication equipment and certain raw materials to support the expansion of domestic manufacturing. Rather than signaling dependence alone, this import intensity reflects active capacity building within a highly specialized semiconductor ecosystem. The overall pattern is one of focused strength in downstream electronics and related materials, positioning South Korea as a structurally concentrated supplier in key segments of quantum-relevant equipment trade.

National Quantum Strategy and Government Support for Commercialization

National quantum strategies differ not just in how much governments spend but in what they are trying to achieve with that spending and how they are structurally organized to achieve it. This section examines the policy intent, funding structures, and specific instruments that each government deploys to support quantum development and accelerate commercial deployment.

United States. U.S. quantum policy rests on a consistent foundational premise: the federal government's role is to sustain the scientific and infrastructure conditions under which private sector–led innovation can thrive rather than to direct which technologies or applications the private sector should pursue. This intent is reflected in the National Quantum Initiative (NQI), originally enacted in 2018. The initiative coordinates quantum R&D across federal agencies while promoting engagement with industry, academia, and national laboratories and advancing foundational research, workforce formation, infrastructure development, and commercialization support.

The NQI formally lapsed in September 2023, and attempts at reauthorization have so far failed to advance through Congress.[14] In January 2026, Senators Maria Cantwell and Todd Young introduced a new bipartisan reauthorization bill, the National Quantum Initiative Reauthorization Act, which would extend the program through December 2034.[15] The proposed legislation reflects a notable shift in emphasis: whereas the original NQI was more oriented toward foundational research, the reauthorization places explicit weight on moving quantum from basic science into practical application, including new testbeds, shared infrastructure, and workforce development pipelines.[16] The bill remains technology-agnostic throughout, avoiding large bets on specific hardware architectures and leaving market forces to determine which approaches succeed.

The scale of U.S. federal funding has grown somewhat since the original NQI was passed. Federal quantum R&D spending increased from $456 million in 2019 to over $1 billion by 2022, and it has remained above $1 billion annually since then.[17] Although these are not trivial sums, in the context of global quantum investment, U.S. federal spending is not the largest.

What distinguishes the U.S. funding environment is what sits alongside federal spending. According to the OECD, U.S. firms capture nearly 60% of all global private quantum investment, even though the country is home to only about 30% of the world's start-ups.[18] In practice, this massive private investment provides a cushion for the U.S. government. Because venture capital is already funding the expensive transition from the lab to the market, federal spending can stay relatively lean without stalling industry commercial development.

This private market is also changing in character. The size of individual deals has gone up significantly for U.S. firms, even though the number of new deals has started to level off. Investors are moving away from spreading small amounts of seed money across many experimental start-ups and are instead concentrating massive checks into a few proven leaders. For example, the size of an average deal for a quantum firm jumped to $51 million in the most recent period, up from approximately $12.5 million just a few years prior.[19] This suggests the U.S. ecosystem is moving from a phase of broad-based discovery into a more intensive industrial scaling phase, where investors are placing high-conviction bets on a limited group of core firms.

Where U.S. policy most directly supports commercialization is through three types of instruments, exemplified by the Defense Advanced Research Projects Agency (DARPA), the Department of Energy, and the QED-C. The first type is performance-gated procurement—government funding tied not to research activity but to commercial results. DARPA's Quantum Benchmarking Initiative selects a group of quantum-computing firms and funds each of them to work toward a specific goal: building a system powerful and reliable enough to solve problems that classical computers cannot by 2033.[20] This is not open-ended research funding. By attaching government money to a defined commercial performance target, DARPA creates an industry-wide milestone that orients private investment and firm strategy toward a concrete finish line rather than a horizon that keeps receding.

The second type is shared infrastructure—facilities that reduce the cost of commercial development for firms that could not sustain them independently. The Department of Energy's network of shared research centers is the primary example.[21] Validating whether a quantum prototype actually works at scale requires highly specialized physical environments: ultra-low temperatures, precision measurement equipment, and shielding from electromagnetic interference that can destabilize quantum systems. Building that infrastructure from scratch is enormously expensive and time-consuming. By providing shared access, the Department of Energy removes a bottleneck that would otherwise force start-ups to spend years and hundreds of millions of dollars just to find out whether their hardware approach is viable. This lowers the cost of the journey from early prototype to deployable product at the stage where commercial development most often stalls. Both this and the DARPA-style procurement instrument share the same underlying logic: the government accelerates commercialization without picking winners, funding the journey and setting the destination while leaving firms to compete on how to get there.

The third type is ecosystem coordination: government-convened bodies that bring together the full range of commercial actors to identify where quantum development is stalling and what needs to happen to move it forward. The QED-C was established directly by Congress through the NQI and is managed by SRI International, a leading nonprofit research institute, with support from the National Institute of Standards and Technology (NIST).[22] It brings together more than 250 member organizations spanning quantum hardware developers, software companies, component suppliers, end users, and academic institutions, with the explicit aim of identifying the gaps—in technology, standards, workforce, and supply chains—that are slowing the journey from laboratory to market. Its work is organized through technical advisory committees covering use cases, enabling technologies, standards and performance metrics, national security, and workforce development, each of which produces roadmaps and conducts pre-competitive research on the specific gaps slowing commercialization. In the U.S. context, where the commercial ecosystem is large, decentralized, and moving across multiple hardware architectures simultaneously, the QED-C's most direct contribution to commercialization is its work on use cases: systematically identifying where quantum sensing, communications, and computing can deliver practical value across industries and then publishing findings that give both developers and potential buyers a clearer picture of where commercial opportunity is actually emerging.[23] However, its role extends beyond mere coordination. The consortium acts as a primary advocacy engine for the industry, translating these identified gaps into specific policy recommendations. Indeed, the QED-C actively shapes the regulatory and legislative environment by testifying on NQI reauthorization, advising on export controls, and coordinating with the NIST on international standards to ensure that U.S. policy proactively supports, rather than unintentionally hinders, commercial scaling.[24]

Coordinating a distributed federal apparatus is a primary challenge for U.S. quantum policy. The National Quantum Coordination Office, housed within the White House Office of Science and Technology Policy, is tasked with synchronizing activities across more than 25 federal departments and agencies.[25] However, because it lacks direct funding authority, its ability to impose a unified strategy on a fragmented landscape is a persistent structural limitation. Investment currently flows through mission-specific agencies with disparate priorities and timelines. Industry groups have flagged this fragmentation as a strategic risk.[26] The Quantum Industry Coalition has urged Congress to swiftly pass the reauthorization of the National Quantum Initiative Act, warning that a lack of sustained, coordinated federal investment could cause the United States to cede its economic and security lead to foreign competitors.[27] While this pluralism allows for multiple simultaneous bets on different technologies, translating coordination intent into action across such a distributed system remains an ongoing challenge for policymakers.

Japan. Japan's quantum policy environment has undergone a fundamental reorientation. Japanese strategy was defined by the 2020 Quantum Technology and Innovation Strategy, which focused almost exclusively on basic research at national institutes such as RIKEN and the University of Tokyo. This earlier model did not have a strong mandate for commercial application. The 2025 Strategy for Quantum Future Industry Development, however, marks a deliberate break from this research-only past. Prime Minister Shigeru Ishiba officially designated 2025 as the "first year of quantum industrialization," shifting the policy's primary metric of success from scientific citations to market creation.[28] The most visible change is the scale of funding: a 1.05 trillion yen ($7.4 billion) package that moves Japan from a marginal spender to a global heavyweight.[29]

Japan's quantum strategy is coordinated across three government bodies with distinct mandates. The Cabinet Office sets the overall national vision and priorities. The Ministry of Education, Culture, Sports, Science and Technology, known as MEXT, is responsible for the foundational scientific pipeline: funding the universities, national laboratories, and long-horizon research programs that produce both the knowledge and the trained researchers the rest of the ecosystem depends on. METI is responsible for pulling that scientific foundation toward commercial application, including industrial deployment, supply chain development, export controls, and international standards.

When it comes to commercialization, Japan's government deploys recognizably similar policy instruments to those of the United States, but the way each one works reflects a fundamentally different industrial context: one where private capital is thin and incumbents dominate. On support for start-ups, Japan has set an explicit policy goal of producing quantum unicorns—start-ups valued at more than $1 billion—and METI has begun providing multibillion-yen investments to promising firms such as OptQC to help them scale toward industrial deployment.[30] This reflects a fundamental difference in how the United States and Japan structure support for emerging quantum firms. The Japanese government acts less like a customer (as DARPA does) and more like a foundational investor, providing the equity injections that the private sector is not yet willing to risk. Whereas U.S. policy is designed to validate a start-up's technology so private investors will fund it, Japanese policy is designed to substitute for those private investors to ensure the existence of the start-up layer absent a comparably deep pool of private funding.

On shared infrastructure, the government funds the Global Research and Development Center for Business by Quantum-AI technology (G-QuAT), housed within Japan's National Institute of Advanced Industrial Science and Technology. G-QuAT provides start-ups and established firms with shared access to superconducting and optical quantum computers. The logic is the same as the Department of Energy's research centers: absorb the cost of specialized testing infrastructure centrally so that individual firms do not have to. The difference is who uses it. In the United States, shared infrastructure primarily serves an independent start-up population. In Japan, it seems to serve a more tightly coupled ecosystem of start-ups and corporate partners working on shared problems.

On ecosystem coordination, Japan's primary vehicle is the Quantum STrategic industry Alliance for Revolution (Q-STAR), a government-organized industry consortium of over 110 members, including Toyota, Mitsubishi, and Fujitsu. Q-STAR and G-QuAT have jointly organized roadmap events focused specifically on the practical application and industrialization of quantum technology, and Q-STAR has launched a start-up accelerator program in partnership with Plug and Play Japan, selecting early-stage firms for structured support toward commercial scale.[31]

China. China's quantum strategy is the global archetype of a "top-down" state-directed model. While the United States relies on a decentralized market, and Japan attempts to engineer one through industrial policy, China treats quantum technology as a critical infrastructure project, akin to its high-speed rail network. The strategy is codified in the 15th Five-Year Plan (2026–30), which officially designates quantum technology as a "new driver of economic growth" and a core pillar of national security.

China has created a massive pool of government funding called the Government Guidance Fund, specifically designed to invest in risky deep-technology companies over very long time horizons without expecting quick returns.[32] This approach is intended to avoid the "valley of death" that kills quantum (and other technology) start-ups elsewhere: the technology takes decades to mature, but most private investors need returns in five to seven years. This fund operates as "patient capital" with a twenty-year lifespan, providing the equity injections necessary to scale domestic firms like QuantumCTek and Origin Quantum without the pressure of quarterly commercial viability.

The government has also established massive regional clusters to co-locate the entire supply chain, most notably in Hefei. The heart of this cluster is Quantum Avenue, a 3.5-kilometer stretch of road where the concentration of start-ups, state laboratories, and industrial parks is the result of a deliberate policy of agglomeration.[33] By bringing all actors—from dilution refrigerator manufacturers to software developers—into the same industrial park, China accelerates the feedback loop between research and production. In Hefei, the provincial government provides direct financial support with a unique risk-tolerance policy: provincial angel funds are permitted to lose up to 40%–50% of their value in pursuit of deep-tech breakthroughs. This provides a safety net for commercial failure that does not exist in the United States or Japan, allowing firms to iterate on hardware like the Zuchongzhi superconducting processors and Jiuzhang photonic systems at a pace dictated by the state, not the market.

China's most powerful commercialization tool is the construction of national infrastructure that creates guaranteed demand for domestic quantum firms. The 15th Five-Year Plan extends this approach to quantum computing by integrating quantum processors into the National Integrated Computing Power Network, a nationwide grid of interconnected supercomputing centers used by government agencies, state enterprises, and universities.[34] Instead of waiting for companies to adopt quantum technologies through normal market channels, China is embedding them directly into infrastructure that already serves millions of users and is mandated to expand.

South Korea. South Korea's quantum policy is best understood as a deliberate catch-up strategy, designed by a government that entered the field late and is candid about that fact. At the launch of South Korea's national quantum strategy, Minister of Science and ICT Lee Jong-ho acknowledged directly: "Although South Korea entered the field of quantum science and technology relatively late, there is still a golden opportunity."[35] The strategy that followed reflects that framing: a mission-based approach to R&D emphasizing defined tasks with sector-specific outcomes, intended to fast-track applied developments in fields like logistics, energy, and materials. This roadmap targets the development of a 1,000-qubit superconducting quantum computer by the early 2030s, the deployment of a 100-kilometer intercity quantum network, and the training of 2,500 core quantum experts and 10,000 quantum-skilled professionals by 2035.

To fuel this catch-up, Seoul has moved from marginal spending to aggressive capital injection. In 2024, the government allocated 128.5 billion won ($96 million) to quantum projects, and the 2025 budget saw a massive 54% increase, reaching 198.4 billion won ($134 million).[36] In 2026, quantum technology has been elevated to a next-generation strategic technology within a broader 5.9 trillion won ministerial portfolio, with a specific focus on the quantum-AI convergence.[37]

South Korea's commercialization model focuses on accelerating start-ups so they can plug into the country's existing industrial giants. The first lever is start-up acceleration. The government's Super-Gap Startup Project, run by the Ministry of SMEs and Startups, allocates roughly $100 million to deep-tech ventures, including quantum companies.[38] Each participating start-up receives a structured five-year support package that includes funding specifically for commercialization activities such as pilot deployments, customer development, and market entry, as well as R&D. By splitting the funding into distinct commercialization and research budgets rather than providing a single undifferentiated grant, the program signals that start-ups are expected to build revenue-generating businesses and not just conduct experiments. The aim is not simply to sustain these firms as research ventures but to integrate them into South Korea's dominant industrial sectors, particularly semiconductors and batteries, so that the start-up layer feeds directly into the country's established industrial base.

The second lever is late-stage scale funding. The Unicorn Bridge Project, launched in January 2026, focuses on identifying and backing the most commercially promising deep-tech start-ups before they reach unicorn status.[39] Rather than distributing funding broadly, the program concentrates resources on a smaller pool of high-potential firms. It selects 50 start-ups to receive subsidies and loan guarantees of up to roughly $14 million each, with second-year funding tied to market validation milestones rather than research outputs.[40] This results-based structure mirrors the logic of DARPA-style programs: funding follows demonstrated commercial progress, not simply ongoing research activity.

The third lever is shared hybrid infrastructure designed to lower the cost of industrial experimentation. The government is building a network of quantum-computing hubs—facilities that link quantum computers with existing classical supercomputing systems so companies can test quantum applications without building their own hardware. The flagship hub connects an imported IonQ trapped-ion quantum computer to the supercomputers at the Korea Institute of Science and Technology Information, a national computing center that already serves universities, research institutes, and industrial firms.[41] This creates a hybrid quantum-classical environment where companies such as Samsung and Hyundai can explore industrial applications at a fraction of the cost of building dedicated quantum facilities. The model resembles the U.S. Department of Energy's quantum research centers and Japan's G-QuAT hubs, but with one key difference: the core hardware is foreign-sourced, reflecting South Korea's continued reliance on international partners for quantum-computing systems while its domestic capabilities mature.

Conclusion

The quantum ecosystems of the United States, Japan, China, and South Korea are not merely at different stages of development on similar trajectories; they are operating on fundamentally different philosophies of innovation. As the transition from laboratory prototypes to industrial deployment accelerates, the "winner" of the quantum race will likely not be the country with the most qubits, but the one whose institutional structure is best adapted to scale, commercialize, and benefit from successes in its specific technological bets.

The United States remains the only ecosystem with the architectural breadth to pursue every hardware modality simultaneously. By leveraging a massive, venture-funded start-up layer and cloud-based on-ramps from platform giants like IBM and Google, the country is best positioned to discover the most valuable applications of quantum computing through sheer market trial and error. However, its reliance on decentralized private capital and a fragmented federal apparatus leaves it vulnerable to funding winters and coordination gaps that state-directed rivals do not face.

Japan's strategy is a lesson in comparative advantage. By over-indexing on ultra-precise timing (sensing) and algorithmic efficiency (software), Japan is positioning itself to be the indispensable provider of the connective tissue for the global quantum economy. Its incumbent-led model is structurally optimized for reliability and integration into existing industrial supply chains.

China has eliminated the "valley of death" by treating quantum as a state-mandated utility. It is poised to scale specific technologies with brute-force efficiency. While this model excels at building massive infrastructure, the recent exit of private tech giants like Alibaba suggests a narrowing of innovation that could struggle to pivot if its primary hardware bets encounter unforeseen technical walls.

South Korea is attempting the world's most aggressive transition from a latecomer to a manufacturing hub. Seoul is realistic about what it can do in the near term and deliberate about building the foundations for a larger role later.

Ultimately, the national ecosystems best positioned for success are those whose institutional structures align with the technological pathways they are pursuing. But as quantum technologies move unevenly from research to commercialization, the features that matter most will shift. A diverse start-up layer, patient state capital, deep incumbent integration, or manufacturing specialization could each prove decisive at different stages. Which of those structural advantages translates most powerfully into sustained leadership as hardware matures, applications consolidate, and markets develop remains an open question. Future analysis will need to examine not only which country builds the strongest systems but which ecosystems have built the kind of structural resilience that allows them to absorb setbacks, capitalize on unexpected breakthroughs, and adapt as the commercial landscape continues to take shape.

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